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THE COPPERBELT UNIVERSITY

MICHAEL CHILUFYA SATA SCHOOL OF MEDICINE

END OF TERM I TEST - MARCH, 2022

MBS 230 - MEDICAL BIOCHEMISTRY I

Student Identification Number (SIN) : _____

Program of Study: _____

MBCbB

Duration of Examination : 02 hours

Total Marks : 80 marks

INSTRUCTIONS TO CANDIDATES

READ CAREFULLY BEFORE YOU START TO ANSWER

- Write your STUDENT IDENTIFICATION NUMBER (SIN) on the space provided on the front page.
- Read the instructions at the beginning of every section (Sections A and B). Answer all questions. No negative marking applies for wrong answers in Section A. Mark allocations for questions in Section B are shown in brackets.
- USE ONLY BLUE OR BLACK INK PEN to answer. Other colours may be used for diagrams only. Answers written in pencil shall not be considered, except for diagrams.
- Answer in this paper only. For section A, circle the letter corresponding to the correct answer and for section B, your answer should be in the provided space. Your answers should be LEGIBLE AND READABLE, otherwise they shall not be considered.
- No rough work is allowed anywhere in the paper, except on the last page i.e specifically allocated space. If any kind of rough work is found on any page, other than on the space specifically allocated, the page (s) shall not be considered.
- No one shall leave the examination room in the first 30 minutes and in the last 30 minutes of examination.

SECTION A: MULTIPLE CHOICE QUESTIONS - CHOOSE ONLY ONE CORRECT AND BEST ANSWER FOR EACH QUESTION AND CIRCLE THE LETTER IN FRONT OF IT. ANSWER ALL QUESTIONS (40 x 1 = 40 MARKS)

1. An amino acid that yields acetoacetyl CoA during catabolism of its carbon skeleton is.....

- a. Glucogenic
- ☒ b. Ketogenic
- c. Hydrophobic
- d. Polar
- e. Polar and hydrophobic

2. is an amino acid that contains an imidazole aromatic ring

- a. Tyrosine
- b. Phenylalanine
- ☒ c. Histidine
- d. Proline
- e. Tryptophan

3. A 50-year-old patient is brought to the emergency department with the diagnosis of myocardial infarction. Which lactate dehydrogenase (LDH) isoenzyme activity would prevail in the patient's serum in the first two days after hospitalisation?

- ☒ a. LDH₁
- b. LDH₂
- c. LDH₃
- d. LDH₄
- e. LDH₅

4. Which of these bonds/forces mainly contribute to stabilisation of secondary structures of proteins?

- a. Hydrophobic bonds
- b. Van der Waal forces
- c. Salt bridges
- ☒ d. Hydrogen bonds
- e. Peptide bonds

5. A chiral carbon of an amino acid arises from the fact that it.....

- a. is symmetric
- b. is bonded to both amino and carboxylic groups
- ☒ c. is bonded to four different chemical groups
- d. has net charge of zero
- e. assumes an L configuration

6. The stability of protein structures is supported by a number of bonds. Which bonds support the primary structure?

- a. Ionic
- ☒ b. Peptide
- c. Hydrogen
- d. Hydrophilic
- e. Hydrophobic

7. The normal hemoglobin of an adult person consists of two identical α and two identical β polypeptide chains. What is this level of structure called?

- a. Primary
- ☒ b. Secondary
- c. Quaternary
- d. Tertiary
- e. Binary

8. Most of the amino acids found in human body are.....

- ☒ a. L-isomers
- b. D-isomers
- c. Optical isomers
- d. D and L- isomers
- e. Optically inactive

9.primarily determines the function of a protein

- a. Number of amino acids
- ☒ b. Spatial conformation
- c. Molecular weight
- d. Chirality of the carbon atoms
- e. Affinity for water

10. Which one of the following amino acids would cause kinks in the peptide backbone, thereby hindering the peptide from forming its usual configuration?

- a. Glycine
- b. Alanine
- c. Glutamic acid
- ☒ d. Proline
- e. Valine

11. The underlying principle of spectrophotometry is.....

- a. The intensity of colour is a measure of the amount of a material in solution
- b. The reflected light is a measure of the amount of a material in solution
- c. The wavelength of the light absorbed is not related to the material in solution
- d. The intensity of light reflected by the solution is indirectly proportional to the light absorbed
- ☒ e. The intensity of light directed to pass through the solution is directly proportional to the colour of the material in solution

12. All of the following statements are correct regarding a peptide, except.....
- a. It contains amino terminus
 - b. It contains carboxyl terminus
 - c. It contains peptide bonds
 - ☒ d. It contains only basic amino acids
 - e. It contains between 2 and 50 amino acids
13. Which of the following is not correct?
- ☒ a. Protoporphyrin is bound to a single iron atom in its Fe^{3+} state
 - b. Iron atom has 6 coordination bonds
 - c. Heme consists of a complex organic ring structure, protoporphyrin
 - d. Heme is found in a number of oxygen transporting proteins
 - e. One of the positions is coordinated to the side chain of a histidine residue
14. Myoglobin is mostly abundant in the.....
- a. Skin
 - b. Bones
 - ☒ c. Muscles
 - d. Nerves
 - e. White blood cells
15. Which of these statements is correct about the role of heme in cytochromes?
- a. Functions as part of the active site
 - b. Binds oxygen irreversibly
 - ☒ c. Functions as an electron carrier
 - d. Functions as a competitive inhibitor in reactions
 - e. Functions as an activator
16. Heme is a complex of.....
- a. protoporphyrin IX and ferric iron
 - ☒ b. protoporphyrin IX and ferrous iron
 - c. porphyrin IX and ferric iron
 - d. protoporphyrin IX and Histidine
 - e. porphyrin IX and ferrous iron
17. is due to the oxidation of heme component to the ferric state
- a. Myoglobin
 - ☒ b. Metmyoglobin
 - c. Hemoglobin C
 - d. Hemoglobin F
 - e. Hemoglobin SC

18. *NADH-Cytochrome b₅ reductase*, an enzyme present in red blood cells, helps to correct the occasional oxidation of the hemoglobin iron. Why are newborns particularly susceptible to methemoglobin-producing compounds?

- a. New borns lack the enzyme
- ☒ b. The capacity of the enzyme is lower
- c. Their hemoglobin lacks the α - globin chain
- d. They only have γ -globin chains
- e. The enzyme at this stage has various mutations

19. Phenylketonuria is a genetic disorder that results from.....

- ☒ a. low or absent activity of an enzyme phenylalanine hydroxylase
- b. low or absent activity of an enzyme tyrosinase
- c. low or absent activity of an enzyme tyrosine oxidase
- d. low or absent activity of an enzyme phenyl oxidase
- e. low or absent activity of an enzyme that converts a non-essential amino acid to an essential amino acid

20. One of the common laboratory experiments in Biochemistry determines electrophoretic spectrum of plasma proteins. What is the physiochemical properties of proteins utilized in this method?

- a. Inability to denaturation
- b. Presence of charge
- ☒ c. Optical activity
- d. Solubility
- e. Viscosity

21. Five isoenzymic forms of lactate dehydrogenase from the human blood serum were isolated and their properties studied. What property indicates that the isoenzymic forms were isolated from the same enzyme?

- ☒ a. Same physiochemical properties
- b. Catalyzation of the same reaction
- c. The same electrophoretic mobility
- d. Tissue localisation
- e. Same molecular weight

22. The portion of a transmembrane protein that is located within the plasma membrane is most likely to be composed of amino acids that are.....

- a. acidic
- b. basic
- ☒ c. hydrophobic
- d. glycosylated
- e. acetylated

23. Which one of the following can influence the charge of amino acid residues in the active site of an enzyme?

- ☒ a. Competitive inhibitor
- ☐ b. Excess substrate
- ☐ c. pH of medium
- ☐ d. Temperature
- ☐ e. Atmospheric pressure

24. Regulation of enzyme activity is the basis for feedback inhibition in enzyme catalysis. Which one of the following represents the most common type of regulation?

- ☐ a. Covalent modification
- ☒ b. Allosteric regulation
- ☐ c. Proteolysis
- ☐ d. Carboxylation
- ☐ e. Hydroxylation

25. Which globin chains are affected in glycosylated Hemoglobin?

- ☐ a. γ -globin chains
- ☐ b. α -globin chains
- ☒ c. β -globin chains
- ☐ d. δ -globin chains
- ☐ e. both γ and α -globin chains

26. Patients, homozygous for hemoglobin C, have a condition that arises from.....

- ☐ a. substitution of lysine by glutamic acid in the β -chain
- ☒ b. substitution of glutamic acid by lysine in the β -chain
- ☐ c. substitution of valine by glutamic acid in the β -chain
- ☐ d. substitution of a polar amino acid by a non-polar one in the β -chain
- ☐ e. substitution of one of the α -chains by a β -chain

27. Which one of these statements is correct about Thalassemias?

- ☒ a. hereditary hemolytic diseases in which there is an imbalance in the synthesis of globin chains
- ☐ b. they are polygenic disorders
- ☐ c. only α -globin chains are affected
- ☐ d. they are not in a class of hemoglobinopathies
- ☐ e. Only γ -chains are affected

28. Which two amino acids in collagen usually undergo post translational modification?

- ☐ a. lysine and histidine
- ☒ b. glycine and alanine
- ☐ c. valine and isoleucine
- ☐ d. lysine and proline
- ☐ e. leucine and glutamic acid

29. Lack of post translational modifications of some amino acids in collagen leads to non-cross linkage of the fibers. Which one of the following refers to the condition that arises?

- a. sickle cell anemia
- b. pernicious anemia
- ☒ c. scurvy
- d. phenylketonuria
- e. hemosiderosis

30. Which one of these statements is correct about the Ehlers-Danlos Syndrome?

- a. results from defective heme synthesis
- b. results from defective porphyrin ring synthesis
- ☒ c. results from defective collagen synthesis
- d. results from defective elastin synthesis
- e. results from defective keratin synthesis

31. is the type of fibrous protein found in hair, nails and outer epidermal layer of mammals

- a. Elastin
- ☒ b. Keratin
- c. Collagen
- d. Procollagen
- e. Proelastin

32. is an example of covalent modification used mainly for regulation of enzyme activity

- ☒ a. Amination
- b. Acetylation
- c. Phosphorylation
- d. Methylation
- e. Decarboxylation

33. Succinyl CoA and are involved in the formation of all the carbon and nitrogen atoms of the porphyrin ring through a condensation reaction catalyzed by an enzyme ALA synthase?

- a. Phenyl alanine
- b. Histidine
- c. Tyrosine
- ☒ d. Glycine
- e. Alanine

34. The catabolism of hemoglobin.....

- a. results in the formation of protoporphyrinogen
- b. results in the formation carbon dioxide
- c. occurs in red blood cells
- d. is the only source of bilirubin
- ☒ e. involves the destruction of the porphyrin ring

35. Which of the following statements is correct about most enzymes?
- a. They increase the activation energy so that the reaction occurs
 - ☒ b. can make a reaction happen faster
 - c. are used once and need to be replaced after catalyzing a reaction
 - d. they are stable in both high acid and high alkaline solutions
 - e. are not specific to the type of substrate they receive
36. Which of the following statements about sickle cell anemia is correct? It is.....
- ☒ a. an autosomal recessive disorder
 - b. an autosomal dominant disorder
 - c. a sex linked disorder
 - d. due to increased solubility of Hemoglobin
 - e. due to reduced deformability of red blood cells
37. Which one of the following statements concerning glutamine is correct?
- ☒ a. Contains an amide group
 - b. Has Glu as its three letter abbreviation
 - c. Classified as an acidic amino acid
 - d. Has a charged polar side chain
 - e. Doesn't have a chiral carbon
38. Which amino acid is capable of forming disulfide bonds?
- a. Glycine
 - b. Proline
 - c. Histidine
 - ☒ d. Cysteine
 - e. Isoleucine
39. Succinate dehydrogenase catalyses the dehydrogenation of succinate to fumarate. Malonic acid is used to inhibit the action of this enzyme. What type of inhibition is this?
- a. Competitive
 - b. Uncompetitive
 - ☒ c. Non competitive
 - d. Feed back
 - e. Allosteric
40. is associated with a condition known as phenylketonuria.
- a. Proline
 - b. Methionine
 - ☒ c. Phenylalanine
 - d. Histidine
 - e. Valine

SECTION B: DESCRIPTIVE ANSWER TYPE/SCENARIO BASED QUESTIONS.
ANSWER ALL QUESTIONS. FOR EACH ANSWER, ILLUSTRATIONS WITH
DIAGRAMS (WHERE POSSIBLE) WILL ATTRACT MAXIMUM MARKS (40
MARKS)

1. Explain how a metabolic pathway can be self-regulating i.e how it can keep itself off and on (1 mark)

Through negative feed back mechanisms, when a reaction is taking place and a lot of products have been formed, the products will change to reactants so as to ~~reduce the~~ regulate the reaction.

2. Explain how denaturation of an enzyme affects its activity. State the effect of the denaturant on each of the structures of the protein enzyme. Do you envisage return of enzyme activity when the denaturant is removed? (2 marks)

Denaturation of an enzyme affects its activity by changing the shape of the active site hence reducing the enzyme activity. In some cases if the denaturant is removed enzymes can resume their activity.

3. State what happens to the rate of product formation in an enzyme catalysed reaction if you continue to:

i. add the substrate (1 mark)

adding a substrate increases the number of products upto a certain point. When all the enzyme active site have been bound adding more substrate will not affect the reaction.

ii add an inhibitor (1 mark)

adding an inhibitor will slow down the rate of enzyme action and product formation. In some cases adding an inhibitor completely stops the reaction.

iii add an enzyme (1 mark)

adding an enzyme increases the rate of the reaction. Enzyme concentration is directly proportional to the number of products formed. So long as we have enough substrate the reaction continues.

4. Describe the following, in reference to proteins (6 marks)

a. Primary structure (1 mark)

Primary structure of proteins is the linear sequence of amino acids held together by peptide bonds.

b. Secondary structure (1 mark)

Secondary structure proteins is the folding of amino acids along its long axis held together by hydrogen bonds.

c. Tertiary structure (1 mark)

Tertiary structure is the three dimensional folding conformational of ~~poly~~ polypeptides.

d. Quaternary structure (1 mark)

Quaternary structure is the assembly of two different polypeptides which can be held together by covalent bonds.

e. What are chaperones and what is their role in protein structure conformation? (2 marks)

Chaperones are proteins which help in the folding or unfolding of other macromolecules. They prevent the formation of aggregates.

5. State two properties of optically active isomers (2 marks)

- i. They rotate plane polarized light either to the left or right
- ii. They are non-super imposable images of each other

6. Explain the statement that an enzyme does not change the position of the equilibrium in a reversible reaction. State three factors that can lead to change in equilibrium position of a reversible reaction (1 mark)

Enzymes do not make a reaction favourable or unfavourable.

7. What is the role of methemoglobin in cyanide poisoning? (2 marks)

Methemoglobin binds to cyanide hence preventing it from inhibiting cytochromes electron carrier mechanism.

8. In patients that have Diabetes mellitus, explain why there is increased amounts of glycosylated hemoglobin (2 marks)

Increased levels of glucose in plasma leads to increased amounts of glycosylated hemoglobin.

Patients that have Diabetes mellitus have high levels of glucose hence an increase in glycosylated hemoglobin.

9. A mutation in a gene that codes for the β chain in hemoglobin causes sickle cell disease. This results in interruption of oxygen flow to tissues. Explain how/why this happens (2 marks)

A mutation in the gene will cause decreased

solubility of oxygen in HbS. The decreased is as a result of the deformed red blood cells that cannot bind oxygen hence affecting the tissues.

10. Explain the relationship that exists between the Michaelis Menten's constant (K_m) and the affinity of the enzyme. (2 marks)

The lower the Michaelis Menten's constant (K_m)

the higher the affinity of an enzyme for a substrate and vice versa.

12. Explain why Sickle cell anaemia is classified as a qualitative hemoglobinopathy (2 marks)

Sickle cell anaemia is classified as qualitative hemoglobinopathy because it results in the mutations of β -globin causing low or decreased affinity of heme for oxygen.

14. Enzymes are often used in clinical diagnosis, however one of the disadvantages to this, for some enzymes, is lack of specificity for a particular tissue. Explain this observation in reference to an anti-TB drug, rifampicin (1 mark)

Enzymes are not specific in that they can bind to a lot of tissues in the body, in the case of anti-TB drug enzymes are not specific.

15. One of the specialised products formed from amino acids, known as a Happy chemical, is Serotonin. It is found in several sites in the body. It is involved in normal and abnormal behaviour and sleep regulation. State the precursor amino acid and the type of reaction(s) that are involved in its formation (1 mark)

Tryptophan
~~Hydroxylation~~ Hydroxylation

16. Histamines are chemical messengers with a wider range of cellular response, among them, allergic, inflammatory. Which amino acid is a precursor for histamines and what type of a reaction takes place to produce histamines? (1 mark)

Histidine
decarboxylation

17. Collagen is the most abundant fibrous protein. Describe its structure, stating the number of polypeptide chains and what variation, in the amino acid sequence, achieves it (2 marks)

Collagen is made up of 3 α polypeptides in a helix arrangement.
Every after 3 amino acids we have glycine

18. Explain an irreversible type of enzyme inhibition. Give examples and state the type of bonds the inhibitor will form with the enzyme (2 marks)

Irreversible type of enzyme inhibition is when the enzymes have been completely denatured. Example: Exposing a baby to some form of chemical radiation makes the baby susceptible to infections.

19. The figure below is a plot showing the effect of substrate concentration on velocity of a reaction. Explain the portions indicated as:

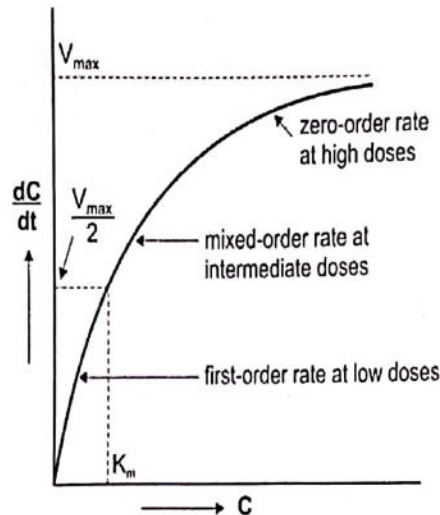


Fig:

i. First order (2 marks)

first order refers to the level at which enzyme active begins.

ii. Mixed order (2 marks)

Mixed order refers to the state at which half of the enzyme active sites have been bound.

iii. Zero order (2 marks)

the level at which all active sites have been bound to.

iv. Define V_{max} (1 mark)

Maximum velocity at which reaction can reach and does not go any further.

v. Define K_m (1 mark)

K_m is defined as the average value for an enzyme substrate reaction.

END OF TEST